Passenger Queue Dynamics and Waiting-Time Simulation for Selected M66 Bus Stops

Project Documentation Submitted to the
Faculty of the School of Computing and Information Technologies

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1. Problem and Objectives

1.1 Project Context

This project focuses on passenger waiting and the use of available bus service across five consecutive eastbound M66 stops in Manhattan. Its real-world concern is whether service available at different times of day is aligned with changing passenger demand along the selected corridor.

The group initially considered studying the EDSA Busway because it is locally relevant to commuters in the Philippines. However, the proposed queue simulation requires compatible stop-level boarding and alighting totals by hour together with a machine-readable bus schedule. During project scoping, the group did not identify a public EDSA source that provided all of these inputs at the required level of detail. The M66 route was therefore selected because the MTA publishes both stop-hour ridership records and static GTFS schedule data. This was a data-feasibility decision and does not imply that the M66 corridor is more important than EDSA.

Passenger waiting on a bus corridor is shaped by two processes that change throughout the day: how quickly passengers arrive and how buses are spaced. When demand rises or buses are unevenly distributed across time, queues can grow between arrivals even when the total number of scheduled trips appears adequate. These effects are difficult to judge from a timetable or aggregate ridership totals alone because queue length and individual waiting time are outcomes of the interaction between passengers and service over time.

The five stops were chosen as one continuous eastbound segment, stop sequences 6 through 10, served by the same full-route M66 pattern. This preserves the physical order of bus and passenger movement while keeping the model feasible within the academic term. The weekday 6:00 to 11:00 AM and 3:00 to 8:00 PM periods were selected as equal five-hour windows for a consistent comparison of contrasting morning and afternoon-to-evening conditions. This controlled scope is not intended to represent the full M66 route or the full service day.

The project is socially relevant because long or unreliable waits reduce the practical accessibility of public transport. It aligns with United Nations Sustainable Development Goal 11, particularly Target 11.2, which calls for safe, affordable, accessible, and sustainable transport systems (United Nations, 2025).

1.2 Problem Statement

The available M66 ridership and schedule records do not directly show how long individual passengers wait, how queues build between buses, how many passengers remain unserved, or how effectively scheduled bus capacity is used. This limits the ability to identify where service and passenger demand are misaligned and whether a feasible redistribution of existing trips could improve passenger outcomes.

1. Unmeasured passenger queue performance. The available records do not reveal the mean and P95 completed waiting time, peak queue, period-end queue, final backlog, or bus occupancy produced by the interaction of passenger arrivals and published bus service at the five selected stops during the two study periods.
2. Vulnerability to specific operating stresses. It is not yet known how a 40 percent increase in passenger demand, the removal of one trip from each of the five study hours, and the

combination of these two conditions would affect waiting time, queue growth, unserved backlog, and bus occupancy relative to normal-demand service.
3. Risk of inefficient hourly service allocation. When a fixed trip budget is concentrated in lower-demand hours while higher-demand hours remain underserved, some buses may be lightly occupied while other passengers experience longer waits or are left behind. A feasible method is needed to redistribute the same scenario-specific number of trips across the five study hours and determine whether passenger outcomes improve without increasing the final backlog.

1.3 Objectives

By the final project submission deadline, the group aims to complete and evaluate a reproducible AI-supported discrete-event simulation for five consecutive eastbound M66 stops that measures queue and waiting-time performance under normal demand, 40 percent higher demand, reduced service of one fewer trip in each study hour, and the combination of higher demand and reduced service, then evaluates feasible hourly trip allocations within each scenario's fixed service budget.

1. Implement and validate separate morning and evening simulations for the five selected stops and estimate mean and P95 completed waiting time, peak queue, period-end queue, final backlog, and bus occupancy using at least 30 independent passenger-arrival replications and 95 percent confidence intervals.
2. Quantify the change from the normal-demand baseline caused by each planned stress condition: 40 percent higher passenger demand; one fewer trip in every study hour, reducing the Morning budget from 43 to 38 trips and the Evening budget from 44 to 39 trips; and the combination of higher demand and reduced service. Report changes in mean and P95 completed waiting time, peak queue, period-end queue, final backlog, and bus occupancy using consistent model rules and replication controls.
3. Use the selected Genetic Algorithm to produce feasible integer trip allocations across the five study hours for the normal-demand baseline and all three stress conditions, compare each selected plan with its matching non-optimized reference using paired replications and 95 percent confidence intervals, and determine whether mean and P95 completed waiting time decrease without an increase in final backlog while reporting the resulting trip plan, candidate headways, peak queue, and changes in mean and maximum bus occupancy.

2. Dataset Analysis

2.1 Dataset Description and Scope

The primary source is the MTA Bus Stop Level Ridership: Beginning 2024 dataset. It reports Date, Hour, Route ID, Direction, Stop ID, Stop Sequence, Boardings, Alightings, and Trips. MTA states that boarding and alighting counts come from Automatic Passenger Counters installed on buses. The MTA overview also notes that not all buses have operational APC coverage and that some records are excluded when APC data are unavailable or counts are zero. The dataset should therefore be treated as APC-observed activity rather than complete passenger demand (MTA, 2026a, 2026b, 2026c).

The supporting source is the current Manhattan static GTFS feed from MTA Developer Resources. The verified feed version is gtfs_m_20260611T155328Z, with feed dates from June 27 to September 5, 2026. The current weekday service used by the project is service_id MQ_C6-Weekday, route_id M66, direction_id 0, trip_headsign EAST SIDE YORK AV CROSSTOWN, and full-route shape_id M660030. This full pattern contains all five modeled stops. During the study hours, 43 eligible full-route trips enter the modeled segment in the morning and 44 in the evening, for 87 trips total (MTA, 2026d; MobilityData, 2026).

The raw project ridership extract keeps all 2025 M66 eastbound stop records because upstream stops are required to estimate how many passengers are already onboard before the first selected stop. The final five-stop analysis dataset retains only weekdays, study hours, and stop sequences 6-10. Because demand is from 2025 and the reference schedule is from 2026, the model evaluates a representative demand pattern against a current service pattern. It is not a reconstruction of a specific historical date.

Table 1. Dataset filtering and retained records

Filtering Stage	Records	Purpose
Full MTA dataset at acquisition	155,387,648	Live source table before project filters.
Calendar year 2025	92,202,449	Website date filter.
2025 + Route M66	165,888	Website route filter.
2025 + M66 + Eastbound	86,959	Downloaded raw project extract.
Weekdays only	64,531	Python weekday filter.
Study hours only	35,372	Hours 6-10 and 15-19.
Selected stop sequences 6-10	12,815	Five-stop corridor.
Final cleaned selected dataset	12,814	One internally inconsistent record removed.

2.2 Key Ridership Variables

Table 2. Key MTA ridership variables

Variable	Meaning	Project Use
Date	Service date of the observation.	Weekday filtering, chronological checks, and missing-combination analysis.
Hour	Hourly interval of the reported metrics.	Study-hour filtering and representative stop-hour demand.
Route ID	Bus route identifier.	Restricts the source data to M66.

Variable	Meaning	Project Use
Direction	Direction of travel.	Restricts the extract to eastbound service.
Stop ID	Historical stop identifier.	Supports source checks, but is not used alone for cross-source joins.
Stop Sequence	Route position of a stop.	Selects the five modeled positions and supports route-consistent mapping.
Boardings	Passengers recorded boarding at the associated stop-hour.	Representative passenger demand used to generate virtual queue arrivals.
Alightings	Passengers recorded exiting at the associated stop-hour.	Reduces bus occupancy before boarding at each stop.
Trips	Buses observed at that route-stop-hour.	Historical service context and data-quality checks. Simulated reference buses come from GTFS.

2.3 Key GTFS Variables

Table 3. Key GTFS variables

Variable	Meaning	Project Use
feed_version	Published GTFS feed identifier.	Documents the exact current feed used.
service_id	Service calendar identifier.	Selects MQ_C6-Weekday after calendar verification.
route_id	Route associated with a trip.	Selects M66.
direction_id	Trip direction.	Selects direction 0, verified as eastbound.
trip_id	Unique scheduled trip identifier.	Represents one bus through the five selected stops.
trip_headsign	Rider-facing trip destination.	Verifies EAST SIDE YORK AV CROSSTOWN.
shape_id	Route pattern identifier.	Selects M660030, the full eastbound pattern.
stop_id / stop_name	Current stop identity and name.	Verifies the physical stop and cross-source mapping.
stop_sequence	Order of stops within a trip.	Selects sequences 6-10 for the modeled segment.

Variable	Meaning	Project Use
arrival_time	Scheduled bus arrival time.	Creates published reference bus-arrival events.
departure_time	Scheduled bus departure time.	Supports schedule verification.

2.4 Data Quality Issues and Treatment

The group profiled the 12,815 selected-stop records before cleaning. There were no missing cells, exact duplicate rows, logical duplicates for the same Date-Hour-Stop Sequence combination, or negative Boardings, Alightings, or Trips values. One record had passenger movement even though Trips was zero. This record was removed because the MTA data dictionary defines Trips as the number of buses serving the route-stop-hour and Alightings as passengers exiting a bus, making the combination internally inconsistent (MTA, 2026c). Data were not removed simply because values were large. Karch (2023) cautions against automatic outlier deletion when an extreme observation may be valid.

Table 4. Data quality issues and treatment

Issue	Extent	Treatment and Basis
Passenger activity with zero Trips	1 selected-stop record; 3 additional upstream records.	Removed only when Boardings or Alightings were positive while Trips was 0. This is internally inconsistent with MTA variable definitions.
Source-absent expected combinations	235 before cleaning; 236 unavailable after the invalid row was removed.	Not filled with zero because MTA states that records may be absent due to APC unavailability or zero APC counts.
Incomplete APC coverage	Dataset-wide limitation.	Treat Boardings and Alightings as APC-observed activity, not complete systemwide demand.
Cross-source Stop ID changes	Sequences 8-10 affected.	Do not join on Stop ID alone. Verify route, direction, stop sequence, and stop name together.
High but plausible counts	Maximum Boardings = 206.	Retained because no rule or source evidence shows the value is invalid.
2025 demand vs 2026 GTFS	Different reference periods.	Interpret results as controlled sensitivity analysis, not historical reconstruction.

The 235 source-absent combinations were spread across 74 dates. Missingness was highest at stop sequence 7 with 79 combinations and during the 7:00-8:00 PM hour with 77 combinations. The absence pattern is therefore not uniform. The group did not impute source-absent combinations as zero because MTA documentation states that records may be absent when APC coverage is unavailable or when APC counts are zero; therefore, absence does not uniquely indicate zero passenger activity (MTA, 2026b). The missingness pattern and the resulting treatment are reported explicitly to preserve transparency in the data-cleaning process, consistent with the reporting principles discussed by Huebner et al. (2020).

Table 5. Verified cross-source stop mapping

Seq.	2025 Ridership Stop ID	Current GTFS Stop ID	Verified Stop Name
6	403492	403492	W 65 St / Central Park West
7	403494	403494	E 65 St / 5 Ave
8	404994	400027	Madison Ave / E 65 St
9	403495	405489	Madison Ave / E 67 St
10	403497 (Jan 1-Mar 28); 405627 (Mar 31-Dec 31)	405627	E 68 St / Lexington Ave

2.5 Dataset Observations and Visualization Insights

Figures 1 through 4 present the stop-level, hour-level, and distributional patterns used to justify separate stop queues, stop-hour-specific passenger inputs, and the testing of alternative hourly bus allocations.

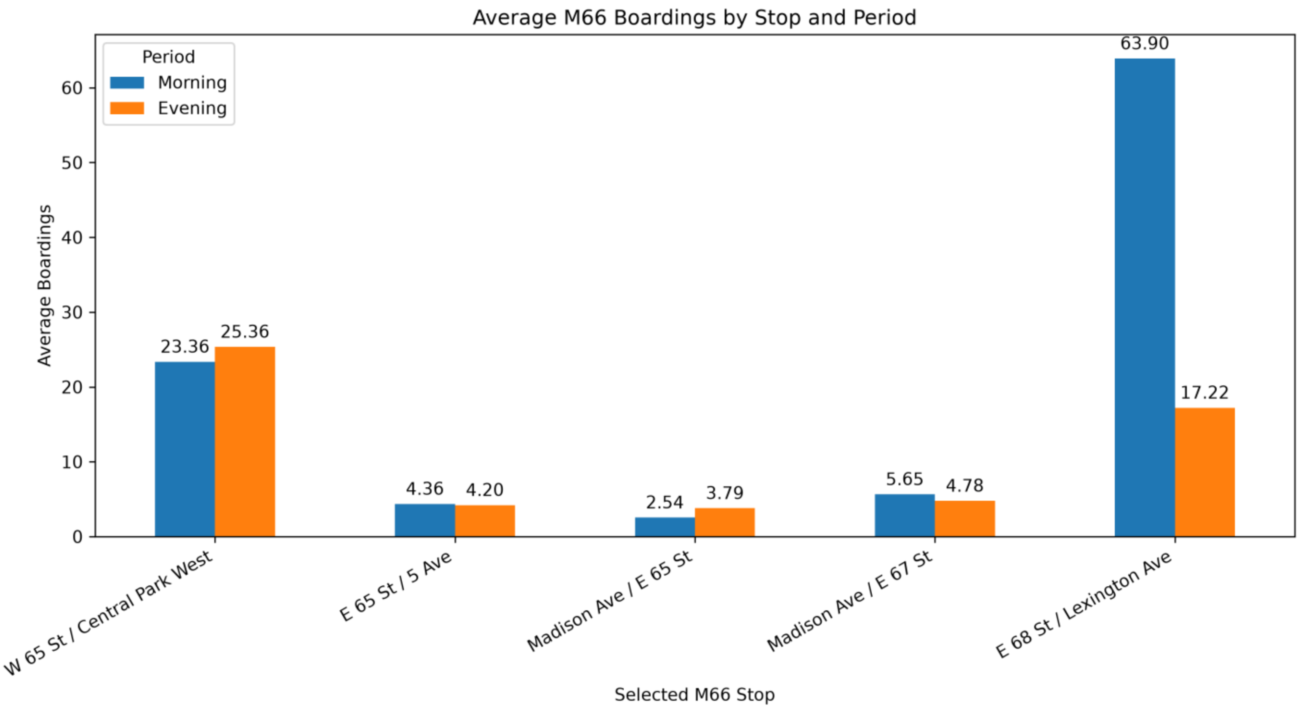

Figure 1. Average M66 Boardings by Stop and Period

Note. Values are mean Boardings per observed stop-hour from the 12,814 cleaned records. The grouped bars separate the Morning and Evening study periods for each selected stop.

Figure 1 shows that representative passenger demand differs substantially across the five selected stops. E 68 St/Lexington Ave has the highest Morning average, while W 65 St/Central Park West remains relatively high in both periods. Across all stops and hours, mean Boardings were 20.11

during the Morning and 11.14 during the Evening, while mean historical Trips recorded in the ridership dataset were similar at 5.01 and 5.03, respectively. These results indicate that the period difference is more strongly associated with passenger demand than with a higher recorded number of bus trips. The stop-level variation also supports modeling the five stops as separate queues rather than as one combined corridor queue.

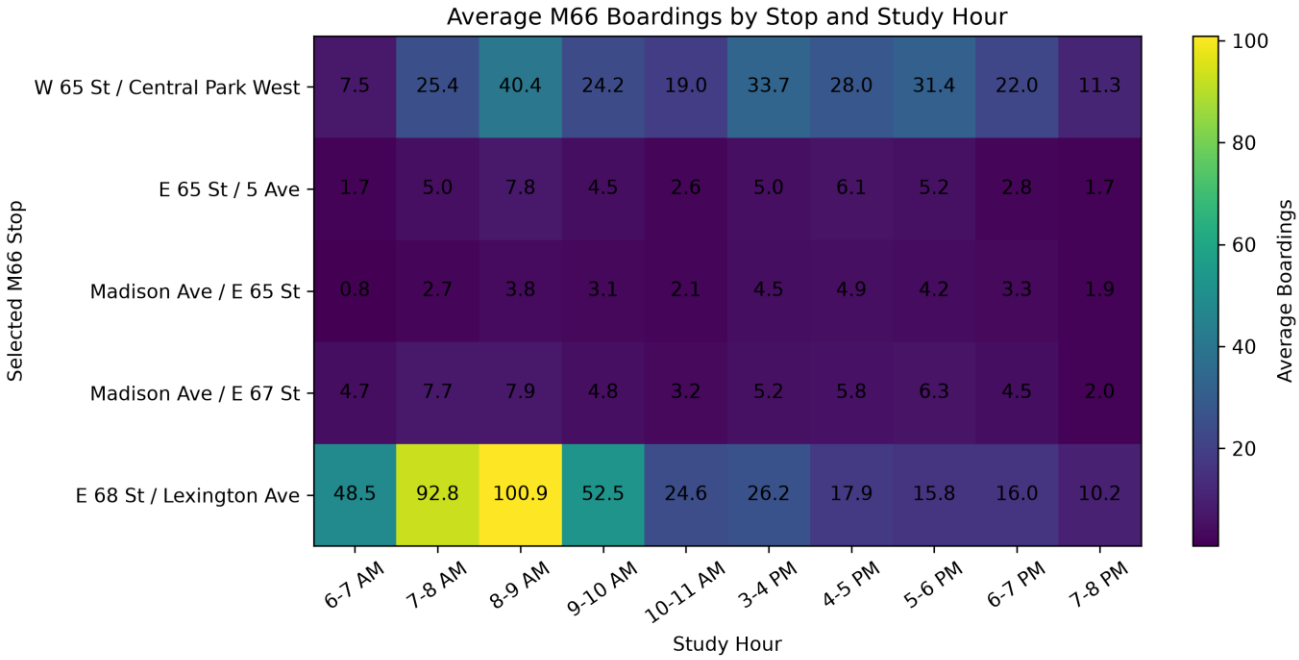

Figure 2. Average M66 Boardings by Selected Stop and Study Hour

Note. Cell values are mean Boardings per observed stop-hour. Greater color intensity represents higher passenger demand.

Figure 2 shows that demand varies by both stop and hour. The highest mean occurs at E 68 St/Lexington Ave from 8:00 to 9:00 AM. Evening demand is highest from 3:00 to 4:00 PM and generally declines during the later study hours. These differences support the use of separate passenger-demand inputs for every selected stop and study hour.

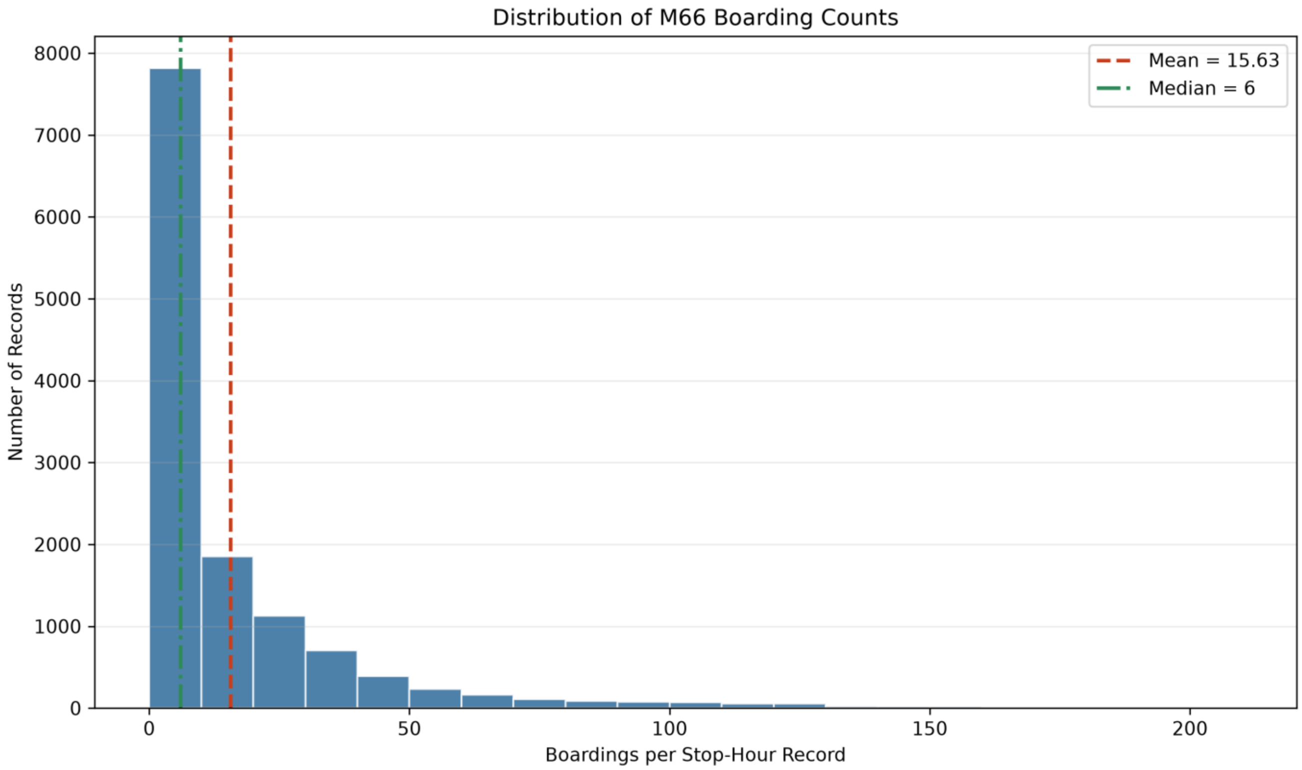

Figure 3. Distribution of M66 Boarding Counts

Note. The histogram represents the 12,814 cleaned stop-hour records. The reference lines show the overall mean of 15.63 and median of 6 Boardings.

Figure 3 shows that Boarding counts are right-skewed and highly variable. Most records are concentrated at relatively low values, while a smaller number extend toward the maximum of 206. The mean of 15.63 is substantially higher than the median of 6, and the standard deviation is 23.24. A single constant passenger-arrival rate would conceal this variation. Therefore, the simulation uses stop-hour-specific demand inputs.

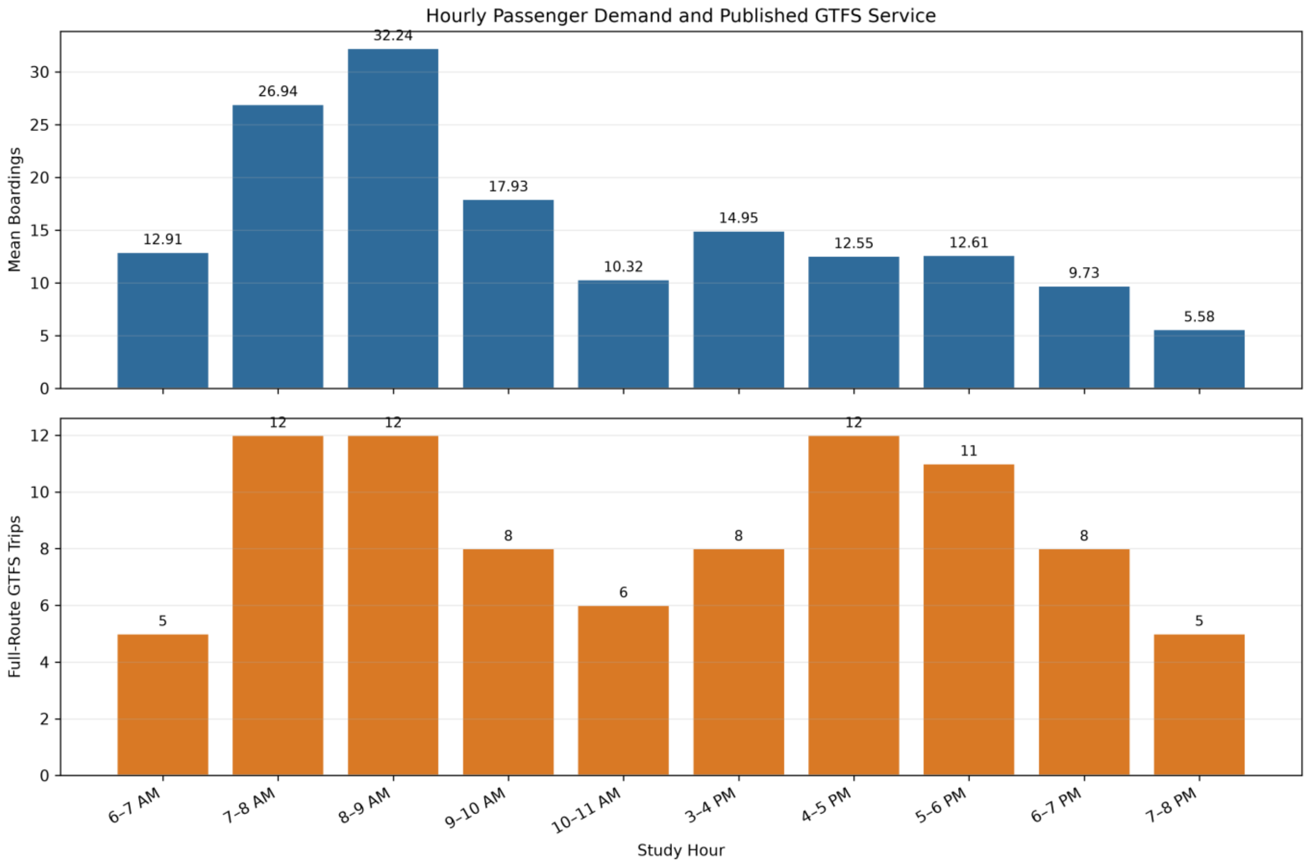

Figure 4. Mean Selected-Stop Boardings and Verified Full-Route GTFS Trips by Study Hour

Note. The upper panel presents mean Boardings from the cleaned 2025 ridership data. The lower panel presents full-route trips from the verified published GTFS schedule. The two panels use different vertical scales because Boardings and trips are different measurements.

Figure 4 shows that representative passenger demand and published GTFS service do not move proportionally in every study hour. From 3:00 to 4:00 PM, mean selected-stop Boardings were 14.95 and the verified schedule contained eight full-route trips. From 4:00 to 5:00 PM, mean Boardings were lower at 12.55, but the schedule contained twelve trips. This observation does not prove that the timetable is inefficient because the project covers only five stops and does not include every operational consideration. However, it establishes a measurable service-allocation pattern that can be tested through simulation and optimization.

2.6 How the Data Support the Simulation and AI Component

The cleaned ridership data provide passenger-demand and alighting inputs for the simulation. Boardings are converted to representative whole-number stop-hour passenger counts using deterministic rounding. Individual virtual passengers are then assigned reproducible arrival times within the corresponding hour. Alightings are also converted to whole-number stop-hour targets and distributed among eligible buses. A separate verified upstream subset based on the physical full-route GTFS stops before sequence 6 is used to estimate the number of passengers already onboard before the first modeled stop.

The GTFS feed provides the published scheduled arrival times used for the non-optimized baseline reference, the hourly trip budgets, the five-stop order, and the time-of-day downstream running-time offsets used to construct candidate schedules. These are scheduled rather than observed real-time arrivals. The AI optimizer does not need a supervised target variable. Instead, it proposes hourly service allocations, and the discrete-event simulation returns the waiting-time and queue metrics used as the optimizer fitness. This makes the cleaned demand data and GTFS schedule jointly sufficient for the selected simulation-based optimization role, while their known limitations remain visible in the model assumptions and interpretation. The dataset-analysis Python notebook and its supporting outputs can be accessed through the GitHub repository link in Appendix B.

3. System, Simulation, and AI Design

3.1 Simulation Question

For the five selected eastbound M66 stops during weekday 6:00 to 11:00 AM and 3:00 to 8:00 PM, how do (1) a 40 percent increase in passenger demand, (2) the removal of one trip from each study hour, and (3) the combination of these conditions change mean and P95 completed waiting time, peak queue, period-end queue, final backlog, and bus occupancy relative to the normal-demand baseline, and can a Genetic Algorithm redistribute the same scenario-specific total number of trips across the five hours to improve mean and P95 completed waiting time without worsening final backlog compared with the matching non-optimized service plan?

3.2 System Representation

The system is a terminating discrete-event queueing model for five consecutive M66 eastbound stops: W 65 St/Central Park West, E 65 St/5 Ave, Madison Ave/E 65 St, Madison Ave/E 67 St, and E 68 St/Lexington Ave. Morning and evening are simulated separately. Passengers are individual virtual entities with a stop and queue-entry time. Buses are scheduled trip entities with an arrival time, current occupancy, assigned alightings, and a fixed capacity of 60 passengers.

Within the normal-demand baseline scenario, the non-optimized reference plan keeps the scheduled arrival times published in the verified static GTFS feed for full-route pattern M660030. These are timetable events, not observed real-time bus arrivals. Keeping the published times unchanged provides a recognizable service benchmark against which the modeled alternatives can be compared. Candidate AI service plans instead use five integer trip counts, one for each study hour in a period. Candidate buses are evenly spaced within each hour using midpoint spacing, then moved through Stops 7 through 10 using the median GTFS travel-time offset for the same study hour. This preserves time-of-day running-time structure while allowing the optimizer to change hourly service allocation. The current implementation excludes the short-turn M660028 pattern because it begins at E 68 St/Lexington Ave and does not serve all five selected stops. This keeps the modeled corridor consistent but may understate total service available specifically at Stop 10.

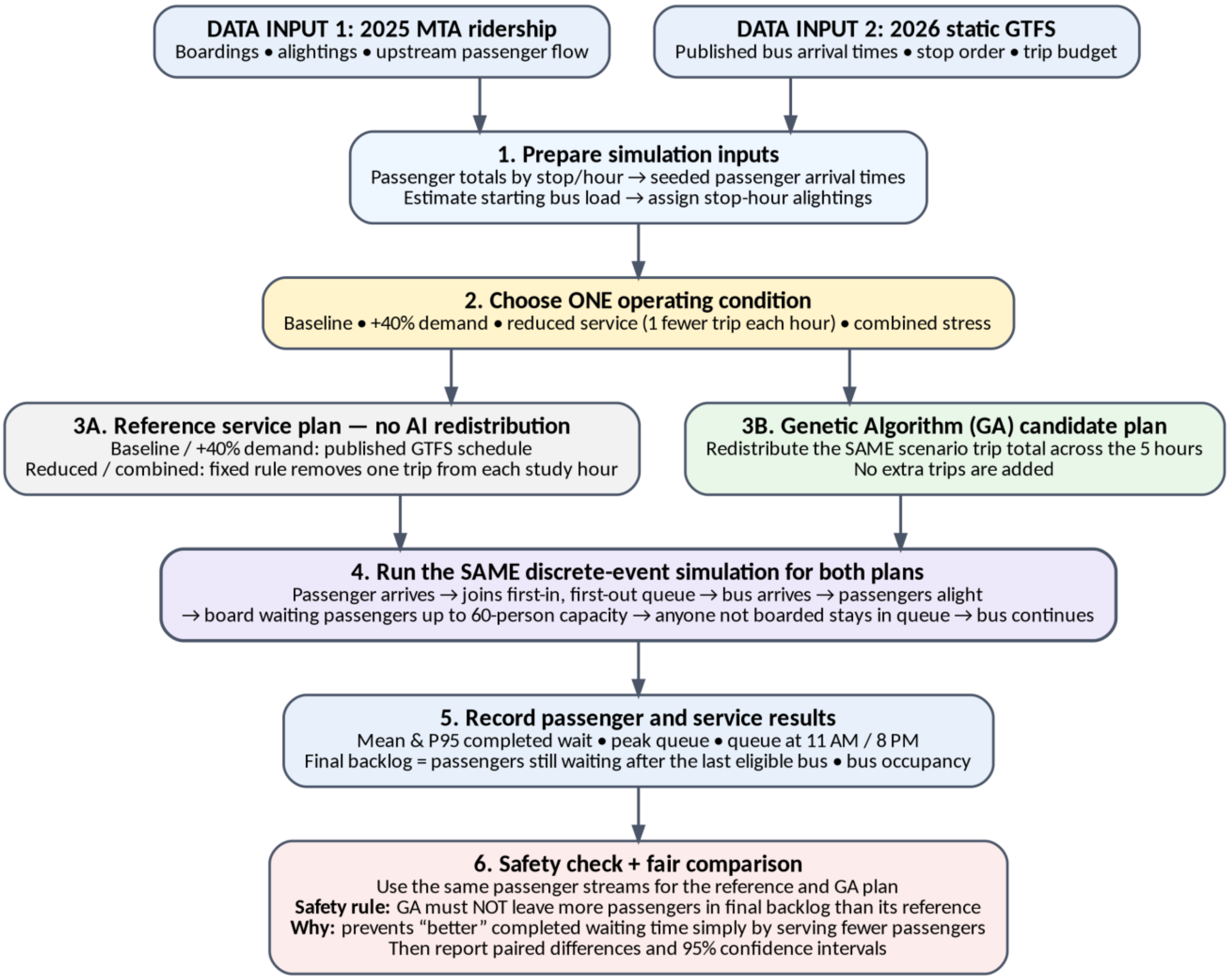

Figure 5. Conceptual framework for the M66 simulation and AI-supported service optimization

Note. Blue boxes show project data and recorded outputs, yellow shows the operating condition and decision checks, gray shows the non-AI reference service, green shows the GA candidate service, and purple shows the shared simulation process. The reference and GA plans are evaluated under the same scenario and passenger streams.

For figure 5, read the diagram from top to bottom. First, the project converts the 2025 ridership data and 2026 GTFS schedule into passenger and bus inputs. Next, one operating condition is chosen. That same condition is tested twice: once using the non-AI reference service plan and once using the Genetic Algorithm (GA) plan. Both plans go through exactly the same passenger-queue and bus-loading simulation. The final safety check prevents the GA from appearing better simply by serving fewer passengers: its final backlog, meaning passengers still waiting after the last eligible bus, cannot be higher than the matching reference plan.

3.3 State Variables

Table 6. Main simulation state variables

State Variable	Description	Type / Initial Value
simulation_clock	Current event time in minutes from midnight.	Float; 360 for morning or 900 for evening.
stop_queues	FIFO queue at each selected stop containing passenger ID and arrival time.	List per stop; empty at period start.
queue_length	Number of passengers currently waiting at a stop.	Integer; 0 at period start.
bus_occupancy	Passengers currently onboard each active trip.	Integer; initialized from upstream estimate.
available_capacity	Remaining boarding space on a bus.	60 minus occupancy.
assigned_alightings	Planned alightings for a trip at a stop.	Integer from stop-hour allocation.
actual_alightings	Realized alightings at the event.	Minimum of assigned alightings and current occupancy.
waiting_records	Boarded passenger ID, arrival time, boarding time, and completed wait.	List of records.
queue_history	Queue length after passenger and bus events.	List of event records.
bus_history	Occupancy, alighting, boarding, and left-waiting information per bus-stop event.	List of event records.
peak_queue / peak_queue_time	Largest observed queue per stop and time of occurrence.	Integer and float.
period_end_queue	Queue immediately after all eligible events at the study-period boundary.	Dictionary by stop.

State Variable	Description	Type / Initial Value
final_backlog	Passengers still waiting after the final eligible bus event.	Dictionary by stop.

3.4 Modeling Assumptions and Basis

Table 7. Modeling assumptions, basis, and possible effects

Assumption	Basis	Possible Effect
The modeled boarding limit is 60 passengers per bus	The project data do not identify the vehicle assigned to each M66 trip or provide vehicle-specific passenger capacities. MTA's Local Bus Schedule Guidelines reported a peak-period service-loading guideline of up to 60 passengers for standard 40-foot buses, while articulated buses had a different guideline (MTA, 2004). The project therefore uses 60 passengers as a transparent modeled boarding limit, not as a claim that every M66 vehicle has an exact physical capacity of 60.	If actual M66 vehicle capacities differ, capacity-limited boarding and backlog can be over- or underestimated.
APC-observed boardings are used as representative passenger-demand input.	The source provides successful stop-hour boardings but does not directly measure everyone who wanted to board. MTA also documents incomplete APC coverage. Boardings are therefore used as a reproducible demand proxy rather than a claim of complete latent demand. This limitation is also consistent with public-transport research showing that observed boarding data may omit passengers who were unable to board because of capacity constraints (Ma et al., 2022).	True demand can be understated when passengers were not observed, were passed by, or chose another option.
Representative stop-hour passenger totals use max(0, floor(p + 0.5)).	The source provides aggregate stop-hour means. Deterministic rounding preserves the intended representative demand level and avoids adding unsupported random variation to the hourly total. In the +40 percent scenarios, the multiplier is applied to the unrounded value before rounding.	Low-volume hours can have small proportional changes because passenger counts must be whole numbers.

Assumption	Basis	Possible Effect
Individual passenger arrival times are uniform within the assigned hour.	Individual passenger arrival times are sampled uniformly within the assigned hour. The source data contain hourly boarding totals rather than individual passenger timestamps. Uniform arrival is a commonly used simplifying assumption in public-transport arrival modeling, although more detailed stochastic processes can better represent within-period fluctuations (Kieu & Cai, 2018). Because the project lacks finer-resolution passenger-arrival data, it operationalizes the hourly totals by assigning reproducible uniform arrival times within each hour.	Within-hour bursts and schedule-aware passenger behavior are not represented.
Morning and Evening begin with empty stop queues.	The source data do not measure passengers already waiting immediately before 6:00 AM or 3:00 PM.	First-hour waiting and queue size may be underestimated if a real queue already existed before the study period.
Passengers board FIFO and do not abandon the queue.	The source data do not contain passenger priority, abandonment, or route-switching behavior. FIFO gives one transparent and reproducible boarding rule.	Long waits and final backlog can be higher than in a real system where some passengers leave or switch routes.
In the non-optimized GTFS reference, buses arrive at their published scheduled times.	No observed real-time M66 arrival feed is included in the project dataset. The reference therefore preserves the verified published timetable exactly.	Traffic delay, early running, bunching, and unplanned cancellation are not represented in the reference.
A bus event at the same timestamp is processed before a passenger event.	This gives an explicit tie rule: a passenger arriving at exactly the same timestamp as a bus does not board that bus.	This affects only exact timestamp ties.
Each eligible full-route GTFS trip_id represents one simulated bus through the five selected stops.	The same trip_id identifies a scheduled trip across Stops 6-10, allowing its occupancy to carry forward consistently from one selected stop to the next.	Vehicle circulation before and after the modeled segment and westbound interactions remain outside the system boundary.

Assumption	Basis	Possible Effect
Initial occupancy is estimated from verified full-route upstream stops 1-5.	For each study hour and service plan, representative upstream net flow is calculated from average upstream boardings minus average upstream alightings. That hourly net flow is divided by the number of eligible buses, rounded to a whole passenger count, and capped to the 0-60 range before each bus reaches Stop 6.	All buses assigned to the same hour receive a representative starting load rather than trip-specific observed occupancy.
Hourly alighting targets are distributed by quotient and remainder across eligible buses.	The representative stop-hour alighting mean is first rounded to a non-negative whole-number target. For target A and n eligible buses, every bus receives floor(A/n) assigned alightings and the first A mod n buses in chronological order receive one additional assigned alighting. Actual alightings are capped by current occupancy.	Trip-specific alighting patterns are simplified, and realized alightings can be lower when a bus does not contain enough passengers.
Source-absent date-stop-hour combinations are not automatically imputed as zero.	MTA documentation states that a record can be absent because APC data were unavailable or because APC counts were zero. Treating every absence as zero would therefore add an unsupported assumption.	Representative demand is based on observed valid records and may still underrepresent conditions with weaker APC coverage.
Candidate buses are evenly spaced within each hour.	The optimizer selects only the number of trips per hour, so an explicit deterministic rule is required to translate each hourly count into exact bus arrival times. Regular headways are relevant to passenger waiting and bus-scheduling performance (Gkiotsalitis, 2020). The midpoint placement itself is a project design choice used to create a reproducible candidate schedule.	Candidate improvement can come from both hourly trip reallocation and within-hour headway regularization.
Candidate downstream times use hour-specific median GTFS offsets from Stop 6.	Published scheduled travel offsets vary by time of day. Using the median offset for the corresponding hour preserves a typical within-hour progression without assuming one overall running time.	Traffic and trip-specific running-time variation are not modeled in candidate schedules.

Assumption	Basis	Possible Effect
The reduced-service reference removes one full-route trip from each study hour using a fixed rule.	For each hour, remove the eligible trip whose Stop 6 arrival is closest to the middle of the hour; if two are equally close, remove the earlier trip. The same trip_id is removed from all five selected stops, and initial occupancy and alightings are recalculated across the remaining buses.	The scenario is reproducible, but results can depend on which specific scheduled trip is removed.
Thirty independent passenger seed sets are the initial replication count, and matched alternatives reuse the same passenger stream.	Independent replications estimate run-to-run uncertainty, while common random numbers reduce noise when comparing a reference plan with its GA alternative (Law, 2024). The precision rule can increase the count in blocks of 10 up to 50.	Additional replications are required if the documented confidence-interval precision rule is not met.
Passenger generation stops at 11:00 AM or 8:00 PM, while already eligible downstream bus events may finish.	The study periods define when new demand enters the model. Buses whose first selected-stop event occurred before the boundary may complete their remaining selected-stop events before final backlog is recorded.	Final backlog represents passengers still waiting after the last eligible modeled bus, not after unlimited later service.

3.5 Simulation Workflow

The simulation first prepares rounded stop-hour passenger totals, exact seeded passenger-arrival times, reference or candidate bus events, capped initial occupancy, and assigned alightings. Morning and Evening are initialized and simulated separately so their demand and service patterns are not mixed.

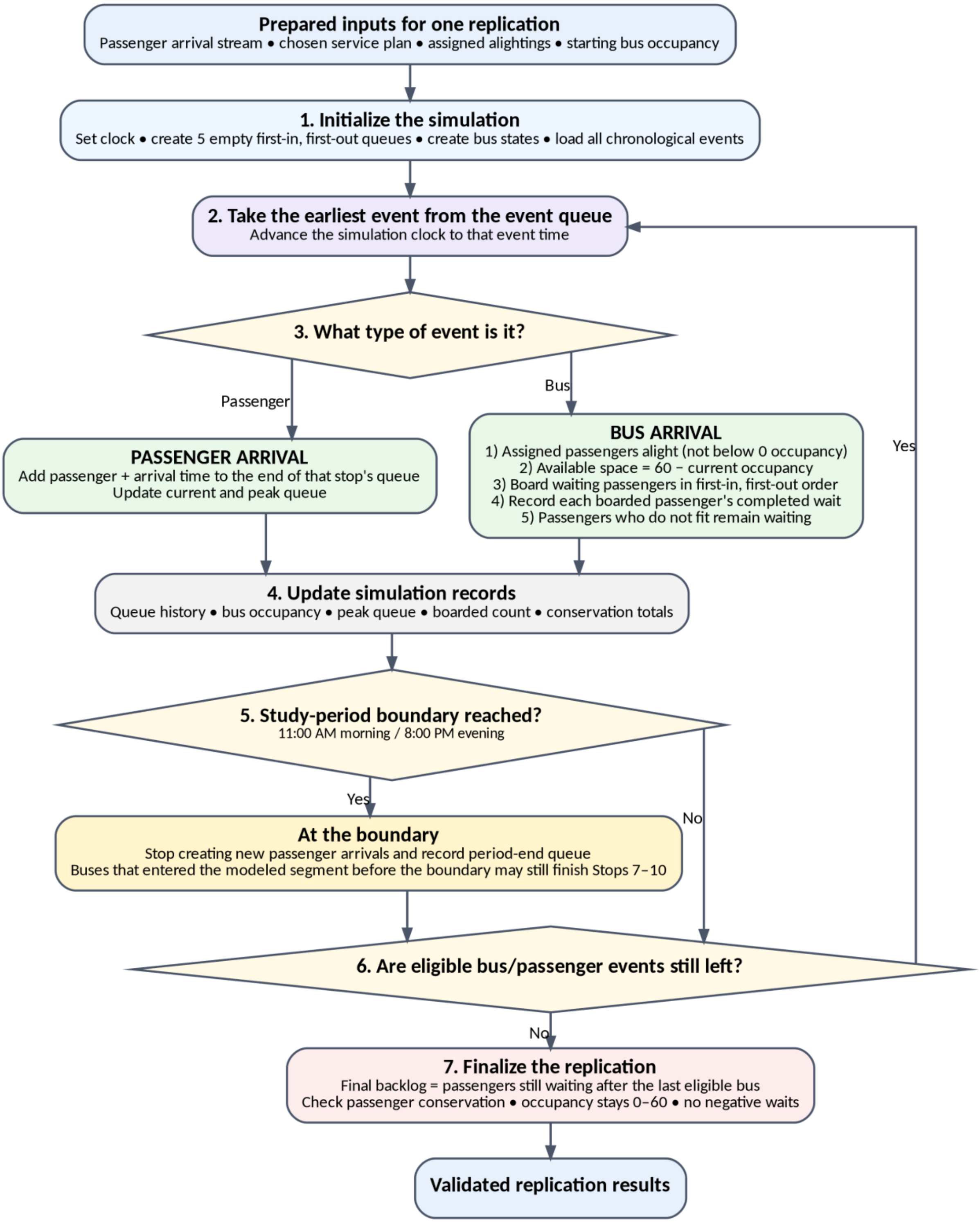

Figure 6. Detailed discrete-event workflow for one M66 simulation replication

Note. The flow shows one complete simulation replication. If a bus and passenger have exactly the same timestamp, the bus event is processed first. Passenger generation ends at 11:00 AM for Morning or 8:00 PM for Evening, while buses that entered the modeled segment before that boundary may finish their remaining selected stops.

For figure 6, read the flow as a repeated event loop. A passenger event adds one passenger to the end of the stop queue. A bus event first removes assigned alighting passengers, calculates the remaining space out of the 60-passenger capacity, and then boards waiting passengers in first-in, first-out order. After every event, the model updates queue and bus records. At the study boundary it records the period-end queue, then allows already eligible buses to finish. Only after the last eligible bus is processed does the model record the final backlog and validate the replication.

3.6 Normal-Demand Baseline Implementation and Verification Status

The normal-demand baseline simulation has already been implemented in Python using the published GTFS schedule as its non-optimized reference plan. The baseline was implemented during the midterm because a working and validated simulation was required to experimentally compare the three candidate AI optimization techniques. GA, SA, and PSO do not evaluate service plans independently; each technique proposes candidate hourly trip allocations, while the same discrete-event simulation measures the resulting passenger waiting time, queues, backlog, and occupancy. The normal-demand baseline therefore served as a controlled test environment for AI-technique screening before the selected technique was applied to the remaining stress scenarios. It generated 500 representative morning passengers and 276 evening passengers in every replication. Thirty independent seed sets were created using NumPy SeedSequence spawning and saved together with all 23,280 individual passenger-event records. The published GTFS reference simulation produced 60 period-level result rows, 30 morning and 30 evening, and every passenger-conservation check passed. No negative completed waiting times or invalid occupancy values were found during the pilot validation. The baseline-simulation Python notebook can be accessed through the GitHub repository link in Appendix B.

Table 8. Preliminary published GTFS reference results across 30 replications

Period	Passengers	Mean Wait	Mean P95 Wait	Mean Peak Queue	Mean Period-End Queue	Mean Final Backlog	Mean Occupancy	Mean Max Occupancy
Evening	276	3.651	7.823	9.867	3.233	3.233	12.217	24.000
Morning	500	3.657	9.081	17.267	3.600	2.167	15.922	29.400

These values verify that the implemented reference model behaves consistently across replications. They are not operational measurements or recommendations; the main analysis uses matched comparisons between each reference service plan and its scenario-specific GA alternative.

3.7 AI Role Rationale: Simulation-Based Service Optimization

The initial system representation proposed passenger-demand prediction as the AI role. After reviewing feedback from Ms. Roselle Gardon on that early design, the group reconsidered whether prediction was the most meaningful use of AI for the project's actual decision problem. The feedback prompted the group to focus on an AI role that directly uses simulation results to support a service decision rather than only estimating an input.

Service optimization was selected because the controllable decision in the model is the allocation of a limited number of bus trips across the five study hours. A metaheuristic optimizer can propose feasible integer service plans, pass each plan to the discrete-event simulation, and use passenger waiting and backlog results as evaluation signals. This structure is consistent with recent public-transport optimization research that treats service frequency or timetable decisions as search problems under operating constraints and passenger-waiting objectives (Ahern et al., 2022; Oliveira et al., 2024).

Table 9. Basis for selecting service optimization as the AI role

Consideration	Rationale
Decision need	The simulation is intended to compare how feasible service allocations affect passenger waiting, queues, backlog, and occupancy.
Feedback	Feedback from Ms. Roselle Gardon on the Initial Project System Representation prompted the group to reconsider whether a prediction-only AI role directly supported the project decision.
Simulation fit	The optimizer can propose hourly trip allocations while the discrete-event simulation evaluates each plan under the same passenger streams and operating rules.
Research basis	Recent bus service and frequency studies use optimization methods to reduce passenger waiting or balance service objectives under operating constraints (Ahern et al., 2022; Oliveira et al., 2024).

3.8 Selected AI Design: Simulation-Based Service Optimization

The selected AI role is an AI-inspired metaheuristic service optimizer. Its decision vector has five integer values, one trip count for each study hour in the selected period. For the normal-demand baseline condition, the total service budget is fixed at the current full-route GTFS total: 43 morning trips or 44 evening trips. During the midterm technique screening, hourly candidate counts were bounded between 5 and 12, the minimum and maximum counts observed in the current study-period GTFS service. The optimizer therefore redistributes existing service rather than receiving more buses automatically. Frequency-setting and bus-scheduling research commonly formulates passenger waiting as an optimization target and includes service-frequency or headway constraints (Oliveira et al., 2024; Gkiotsalitis, 2020). The same 5-to-12 trips-per-hour bounds will be retained when GA is applied to the baseline and three final stress scenarios; only the scenario-specific total trip budget changes.

The common fitness function is the mean completed passenger waiting time across matched simulation replications. Lower is better. Final backlog is used as a feasibility safeguard so a plan cannot appear better simply because more passengers remain unserved and therefore do not contribute a completed waiting time. Peak queue, P95 waiting time, and occupancy are retained as secondary evaluation measures rather than combined using arbitrary weights.

Table 10. AI service-optimization design

Design Element	Specification
Decision vector	Five integer hourly trip counts for Morning or Evening.
Normal-demand budget	Morning = 43 total trips; Evening = 44 total trips.

Design Element	Specification
Hourly trip bounds	5 to 12 trips per hour, derived from the minimum and maximum hourly counts in the current GTFS study-period service and retained for the final scenario optimization.
Candidate Stop 6 times	Evenly spaced at the midpoint of equal headway intervals within each hour.
Candidate downstream times	Stop 6 arrival plus the median GTFS offset for the same hour and downstream stop.
Primary fitness	Mean completed waiting time across matched simulation replications. Lower is better, but only plans that pass the backlog safety check are eligible.
Backlog safety check	Mean final backlog cannot exceed the matching non-optimized reference. This prevents the optimizer from looking better simply by leaving more passengers unserved.
Secondary metrics	P95 completed wait, peak queue, final backlog, bus occupancy.

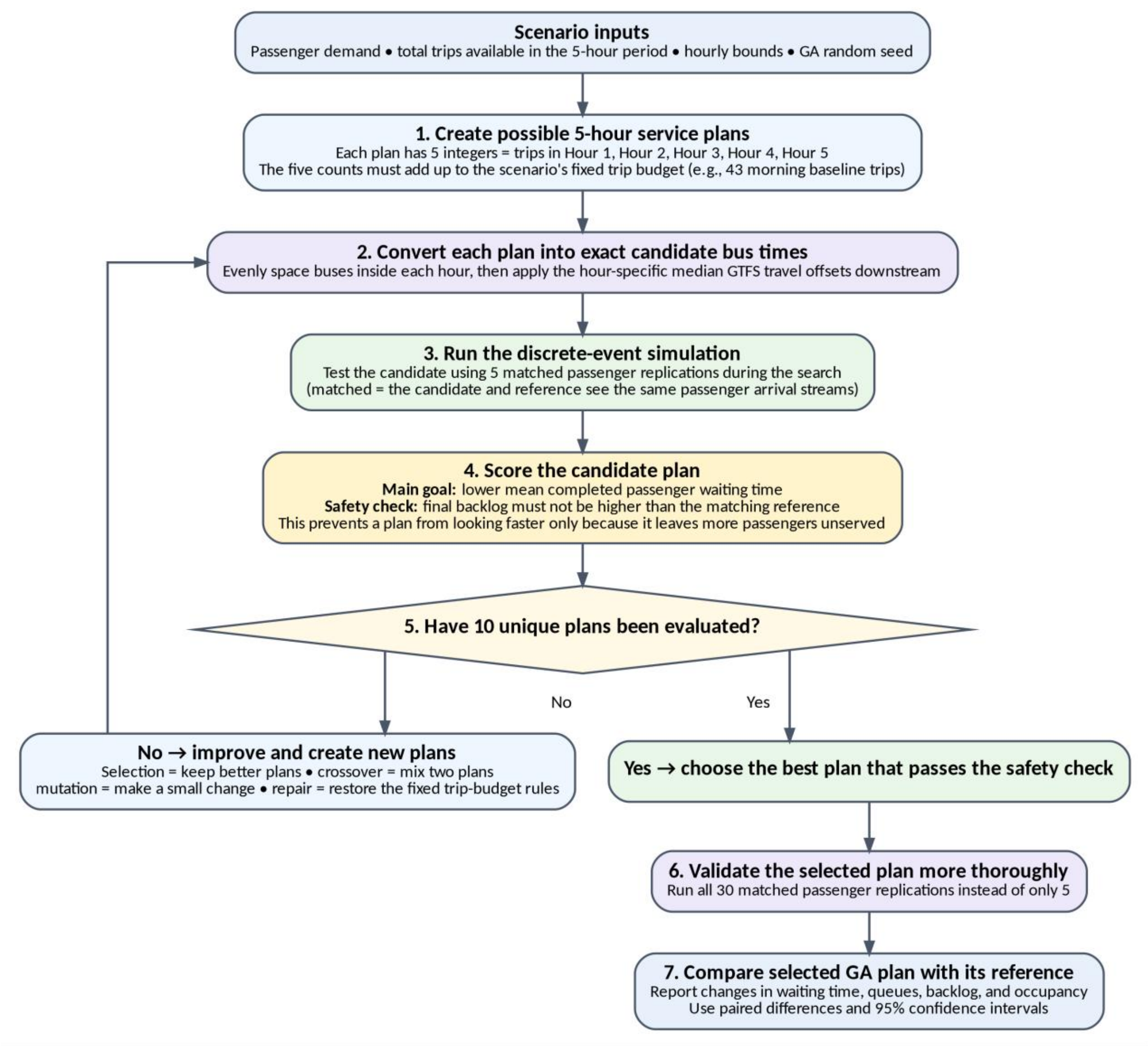

Figure 7. Genetic Algorithm service-optimization and evaluation loop

Note. During the GA search, each candidate plan is tested on five matched passenger replications to keep runtime manageable. "Matched" means the candidate and its reference are evaluated using the same passenger-arrival streams. After the search, the selected plan is re-evaluated using all 30 replications.

Figure 7 shows the Genetic Algorithm searches for a better way to distribute the available trips across the five study hours. Each candidate is first converted into exact bus arrival times and then tested using the discrete-event simulation. The main score is mean completed passenger waiting time. However, a candidate is allowed to compete only if it does not leave more passengers in the final backlog than the matching reference. If fewer than 10 unique plans have been evaluated, the GA keeps better plans, combines them, makes small changes, repairs any invalid trip totals, and tests again. The best plan that passes the safety check is then validated using all 30 matched replications. If the initially selected plan fails the backlog safeguard during the full 30-replication validation, the next-best evaluated plan that satisfies the safeguard under all 30 matched replications will be selected. If no evaluated plan satisfies the safeguard, the scenario will be reported as having no acceptable optimized improvement under the tested search budget.

3.9 Experimental Comparison of Three AI Techniques

Three AI-inspired metaheuristics were tested because each can search a discrete, non-linear service-allocation space without requiring a differentiable simulation objective. Genetic Algorithm is supported by recent bus frequency-setting research that directly minimizes passenger waiting time (Oliveira et al., 2024). Simulated Annealing has been used for integrated bus route and service-frequency optimization (Ahern et al., 2022). Particle Swarm Optimization has been applied to bus timetable and headway optimization to reduce waiting and system cost (Han et al., 2022).

The midterm comparison used the same constraints, candidate-schedule generator, simulation engine, and fitness rule for all techniques. To keep search time manageable, each optimizer run had an equal budget of 10 unique service plans and used the first five matched passenger replications during search. The best plan from each run was then validated using all 30 passenger replications. Each technique was repeated with optimizer seeds 101, 202, and 303 for both Morning and Evening. This is a controlled midterm screening, not an exhaustive parameter-tuning study. The AI-technique-selection Python code and exported comparison results can be accessed through the GitHub repository link in Appendix B.

Table 11. Overall AI technique screening results

Technique	Average Validated Mean Best Wait (min)	Average Run-to-Run SD
GA	3.525	0.019
PSO	3.541	0.037
SA	3.553	0.039

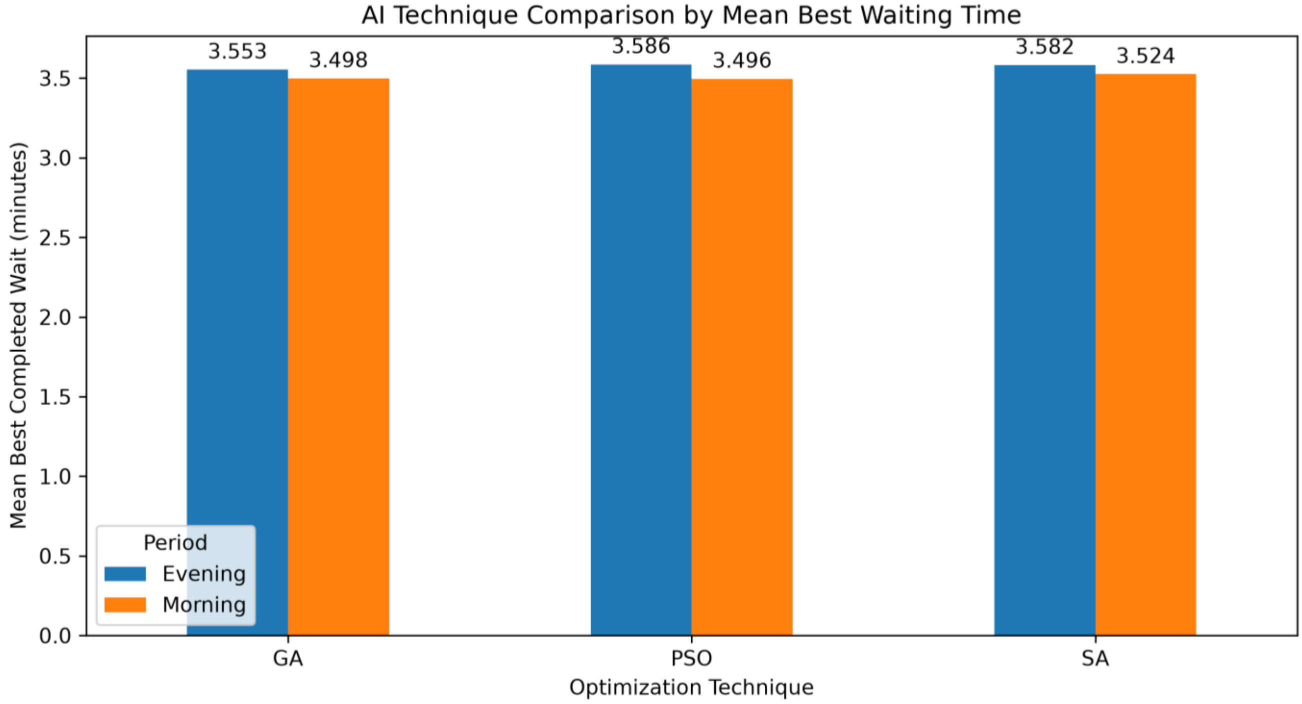

Figure 8. AI technique comparison by average validated mean best waiting time

Note. Lower values are better. GA also had the lowest average run-to-run standard deviation in the controlled midterm screening.

Figure 8 compares the best validated waiting-time performance produced by GA, PSO, and SA during the controlled midterm screening. Lower values are better, and the result supports retaining GA because it achieved the lowest overall validated mean best waiting time while also showing the lowest run-to-run variation.

Genetic Algorithm was selected for the final project because it had the lowest overall average validated best waiting time at 3.525 minutes and the lowest average run-to-run standard deviation at 0.019. PSO was second at 3.541 minutes with SD 0.037, and SA was third at 3.553 minutes with SD 0.039. The selection is based on the group's controlled midterm experiment. GA will remain the selected technique for the final scenarios unless a verified implementation error is identified or the final scenario constraints make the current GA implementation infeasible.

3.10 Preliminary Selected GA Service Plans

Table 12. Selected GA normal-demand service plans from midterm screening

Period	Selected GA Plan	Mean Wait	Mean Backlog	Mean Peak Queue	Mean P95 Wait
Morning	(8, 9, 11, 9, 6)	3.488	1.833	17.500	7.146
Evening	(11, 7, 9, 10, 7)	3.532	0.967	8.633	7.473

For each period, the selected GA plan was the feasible plan with the lowest 30-replication validated mean completed waiting time among the plans produced using optimizer seeds 101, 202, and 303. The selected Morning plan is (8, 9, 11, 9, 6), which still totals 43 trips. The selected Evening plan is (11, 7, 9, 10, 7), which still totals 44 trips. These are simulation candidates, not recommendations for MTA implementation. Candidate timing is synthetic and evenly spaced within each hour, so observed improvement can come from both hourly reallocation and headway regularization.

3.11 Expected Outputs

The final implementation will produce four groups of outputs that directly answer the simulation question and project objectives:

- **Baseline performance results** showing mean and P95 completed waiting time, peak queue, period-end queue, final backlog, and bus occupancy for the Morning and Evening periods. Stop-level or stop-hour results will identify where queue pressure is concentrated.
- **Stress-scenario comparisons** showing how a 40 percent increase in passenger demand, the removal of one trip from each study hour, and the combination of these conditions change the defined performance metrics relative to normal-demand service. Comparison tables and selected queue-pressure visualizations will summarize the results without producing repetitive figures for every replication.
- **GA service-allocation results** showing the selected number of trips assigned to each study hour under the baseline and three stress scenarios. Each GA-optimized plan will be compared with its corresponding non-optimized reference plan using paired differences and 95 percent confidence intervals. The results will be presented as simulation-based decision support rather than direct operational recommendations.
- **Validation and reproducibility evidence** confirming passenger conservation, non-negative waiting times, valid bus occupancy, replication precision, and fixed random seeds. The project

repository will contain the model settings, scenario inputs, selected plans, replication results, and Python code used to generate the reported tables and figures.

4. Experimental and Evaluation Plan

4.1 Experimental Logic

The experimental plan answers the simulation question in two stages. First, it measures how 40 percent higher passenger demand, one fewer trip in each study hour, and the combination of these conditions change the primary passenger and service metrics relative to the normal-demand baseline. Second, within each scenario, it compares the selected GA allocation with the matching non-optimized service plan using paired replications and confidence intervals. This separates the effect of operating stress from the effect of hourly service reallocation.

4.2 Baseline Scenario and Three Experimental Scenarios

The baseline scenario uses normal representative demand and the published GTFS schedule as its non-optimized reference. Each stress scenario also has a matching non-optimized service plan under the same passenger-demand and total-service condition. The GA alternative is compared with that matching reference because an optimized result alone cannot show whether the service allocation improves performance. The reference and GA alternative are therefore two service plans evaluated within each scenario, not additional scenarios.

Table 13. Baseline scenario and three experimental scenarios

Scenario	Passenger Demand	Non-Optimized Reference Service	GA-Optimized Alternative	Purpose
Baseline: Normal demand	Normal representative demand.	Published GTFS scheduled arrivals; 43 Morning and 44 Evening full-route trips.	Reallocates the same 43/44 trip budgets and generates candidate times.	Tests whether allocation and regularity can reduce waiting without adding trips.
Scenario 1: Elevated demand	Each unrounded stop-hour demand p becomes max(0, floor(1.40p + 0.5)).	Same published GTFS schedule and 43/44 total trip budgets.	Reallocates the 43/44 trip budgets under higher demand.	Tests resilience to a substantial demand shock.
Scenario 2: Reduced service	Normal representative demand.	One full-route trip is removed from each study hour. Remove the trip whose Stop 6 arrival is closest to the hour midpoint; a tie removes the earlier trip. Remove the same trip_id at all five stops. Morning = 38 trips; Evening = 39 trips.	Reallocates the reduced 38/39 trip budgets.	Tests a sustained service shortfall.

Scenario	Passenger Demand	Non-Optimized Reference Service	GA-Optimized Alternative	Purpose
Scenario 3: Combined stress	40 percent higher demand using the same pre-rounding rule.	Same reduced-service reference; 38 Morning and 39 Evening trips.	Reallocates the reduced 38/39 budgets under higher demand.	Tests the most difficult planned condition.

The 40 percent demand increase is a stress-test level, not a forecast. It is intentionally large enough to test system sensitivity but still within the same order of magnitude as the strong demand differences already observed across M66 stops and periods. Recent transport simulation work has also used a 40 percent traffic-demand increase as a sensitivity condition, which provides a useful precedent for treating 40 percent as a stress level rather than a prediction (Olstam et al., 2025).

The reduced-service budget is based on removing one full-route trip from each of the five study hours before optimization. Within each hour, the matching non-optimized reference removes the trip whose Stop 6 arrival is closest to the middle of the hour; if two trips are equally close, the earlier trip is removed, and the same trip_id is removed from all five selected stops. Initial occupancy and representative alightings are then recalculated across the remaining buses. This reduces the period total from 43 to 38 in the morning, an 11.6 percent reduction, and from 44 to 39 in the evening, an 11.4 percent reduction. One trip per hour is the smallest uniform whole-trip reduction that creates a sustained shortfall across all five study hours. The GA may then redistribute the resulting period budget instead of being forced to keep one fewer trip in every individual hour.

4.3 Passenger Streams and Common Random Numbers

The baseline implementation stores 30 reproducible passenger streams, one for each seed set. Each stream contains the same rounded stop-hour totals but different uniformly generated arrival times. A reference service plan and its GA alternative reuse the same passenger stream for the same replication. For the elevated-demand scenarios, the final implementation will preserve the baseline passengers and add only the extra passengers needed to reach the 40 percent demand target. This keeps the common part of demand matched across scenarios and makes paired differences easier to interpret. Law (2024) recommends common random numbers as a variance-reduction method when comparing alternatives.

4.4 Replication Count and Precision Rule

The simulation starts with 30 independent seed sets for each period. Replication-level outputs are summarized with t-based 95 percent confidence intervals. Following the procedure already defined in Project Deliverable 1, if the confidence-interval half-width for mean completed wait or peak queue is greater than 10 percent of the mean, the group will add replications in blocks of 10 up to 50. When the mean is below 10 minutes or 10 passengers, the required absolute half-width is 1 minute or 1 passenger. Hoad et al. (2007) support choosing replication counts based on required output precision instead of using a fixed number without checking accuracy.

Table 14. Baseline replication precision check

Period	Metric	n	Mean	95% CI Low	95% CI High	Half-Width	Required	Passed
Morning	Mean Completed Wait	30	3.657	3.616	3.698	0.041	1.000	Yes
Morning	Peak Queue	30	17.267	16.520	18.013	0.746	1.727	Yes
Evening	Mean Completed Wait	30	3.651	3.590	3.712	0.061	1.000	Yes
Evening	Peak Queue	30	9.867	9.280	10.453	0.586	1.000	Yes

All four normal-demand GTFS reference precision checks passed at 30 replications, so 40 or 50 runs were not required for the current reference experiment. The same rule will be applied to final scenario outputs when needed.

4.5 Simulation Evaluation Metrics

Table 15. Simulation evaluation metrics

Metric	Definition	Why It Matters
Mean completed wait	Average boarding time minus queue-entry time for passengers who board.	Primary passenger experience measure and AI fitness.
P95 completed wait	95th percentile of completed waiting time.	Shows long-tail waiting that the mean can hide.
Peak queue	Largest number of waiting passengers observed at any selected stop.	Shows the maximum queue pressure.
Period-end queue	Passengers waiting at 11:00 AM or 8:00 PM after boundary events are processed.	Shows unresolved demand at the end of passenger generation.
Final backlog	Passengers still waiting after the final eligible bus finishes the modeled segment.	Prevents the optimizer from appearing better by leaving more passengers unserved.
Mean and maximum bus occupancy	Passengers onboard after bus-stop events.	Checks capacity use and crowding pressure.
Passenger conservation	Generated passengers = completed boarders + final backlog.	Core internal validation rule.

4.6 AI Evaluation Metrics and Technique Selection Criteria

Table 16. AI evaluation metrics and controls

AI Evaluation Measure	Use
Validated mean best waiting time	Primary technique-selection metric. Each run's best plan is re-evaluated using all 30 passenger replications.
Run-to-run standard deviation	Measures optimizer stability across seeds 101, 202, and 303. Lower variation is preferred when primary performance is similar.
Feasibility rate	Checks service-budget constraints, backlog safeguard, passenger conservation, and capacity validity.
Validated mean backlog	Ensures lower completed waiting time is not achieved by leaving more passengers unserved.
Validated mean P95 wait and peak queue	Secondary passenger-service measures used to detect trade-offs.
Equal evaluation budget	All three techniques receive the same 10 unique-plan search budget in the midterm screening.

4.7 Criteria for Comparing Final Scenario Results

1. Use matched replication numbers and the same passenger streams for each reference service plan and its GA alternative.
2. Calculate the paired difference as optimized result minus reference result for each replication. Negative waiting-time and backlog differences indicate improvement.
3. Report the mean paired difference and a 95 percent confidence interval. A waiting-time improvement is considered statistically supported when the 95 percent interval is entirely below zero.
4. Require passenger conservation and no occupancy outside 0-60 for every run.
5. Apply the backlog feasibility safeguard relative to the matching scenario reference before treating a candidate plan as acceptable.
6. Interpret P95 wait, peak queue, final backlog, and bus occupancy together. A plan should not be described as universally better if lower mean waiting time is accompanied by substantially worse queueing, backlog, or crowding-related occupancy results.

4.8 Preliminary Midterm Comparison: Selected GA Plans vs Published GTFS Reference

Table 17. Preliminary paired comparison of the published GTFS reference and selected GA normal-demand plans

Period	Metric	GTFS Reference Mean	Selected GA Mean	Paired Difference	95% CI of Difference
Morning	Mean Completed Wait	3.657	3.488	-0.169	[-0.220, -0.118]
Morning	P95 Completed Wait	9.081	7.146	-1.935	[-2.056, -1.813]

Period	Metric	GTFS Reference Mean	Selected GA Mean	Paired Difference	95% CI of Difference
Morning	Peak Queue	17.267	17.500	0.233	[-0.808, 1.274]
Morning	Final Backlog	2.167	1.833	-0.333	[-0.537, -0.129]
Evening	Mean Completed Wait	3.651	3.532	-0.119	[-0.195, -0.043]
Evening	P95 Completed Wait	7.823	7.473	-0.350	[-0.556, -0.145]
Evening	Peak Queue	9.867	8.633	-1.233	[-1.819, -0.648]
Evening	Final Backlog	3.233	0.967	-2.267	[-2.924, -1.609]

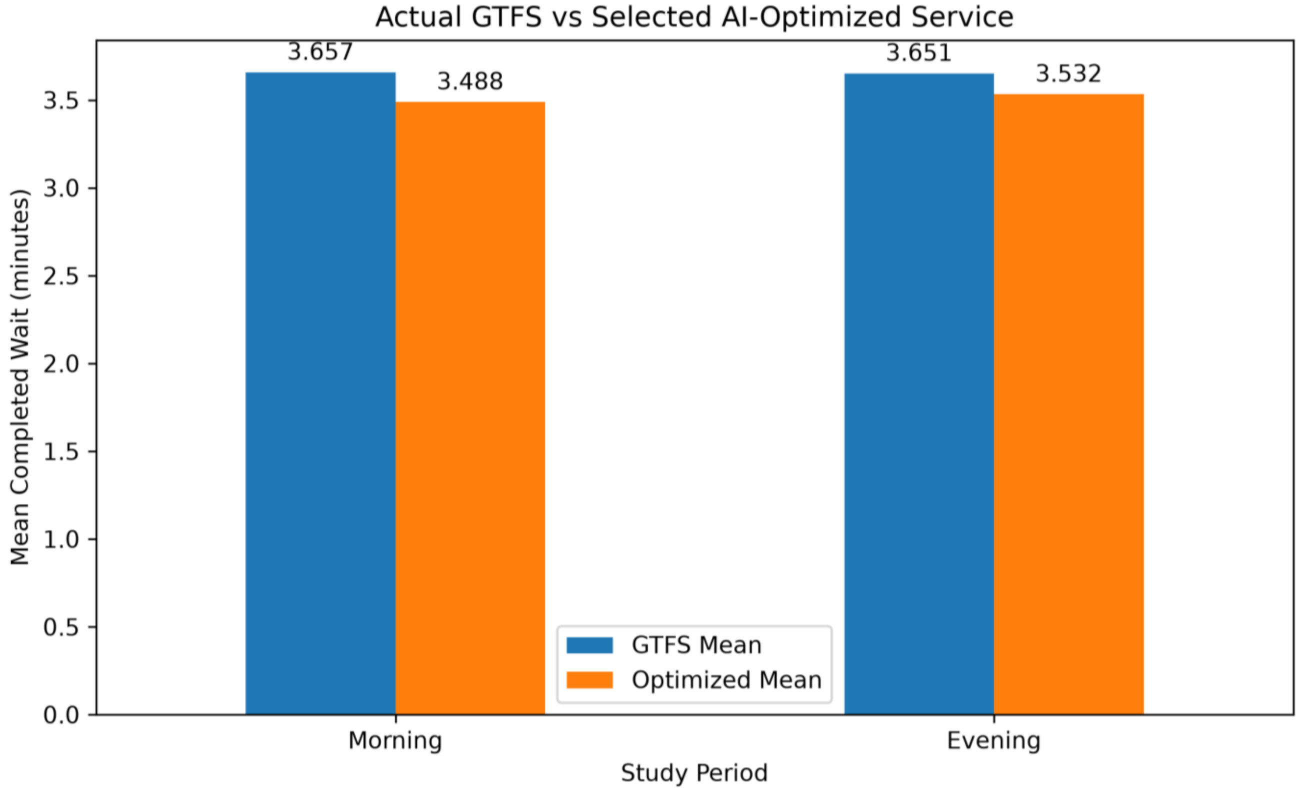

Figure 9. Published GTFS reference versus selected GA plan for mean completed waiting time

Note. The selected GA plans use the same total number of trips as the published GTFS reference. These are preliminary simulation results, not observed M66 performance or operational recommendations.

Figure 9 compares the published GTFS reference with the selected normal-demand GA plan using the same total number of trips. Blue represents the published GTFS reference and orange represents the selected GA plan. The labels above the bars show that the GA plan has a 4.6 percent lower mean completed wait in the Morning and a 3.3 percent lower mean completed wait in the Evening in this preliminary comparison. These differences come from modeled service redistribution and regularity, not from adding more trips.

The preliminary selected GA plan reduced mean completed waiting time by about 0.169 minute in the morning, or 4.6 percent, and by about 0.119 minute in the evening, or 3.3 percent. The paired 95 percent intervals for mean wait were below zero in both periods. Morning peak queue was 0.233

passenger higher on average and its confidence interval crossed zero, showing that not every metric improved. Evening peak queue and final backlog both improved. These findings confirm that the evaluation framework can detect trade-offs and support paired statistical comparison. They do not yet answer the final high-demand, reduced-service, or combined-scenario questions.

The baseline outputs in Tables 8, 14, and 17 and Figure 9 are preliminary midterm evidence. The baseline discrete-event simulation was implemented at midterm for two purposes: first, to verify the event logic, passenger conservation, capacity limits, and replication precision; and second, to provide one common simulation evaluator for the controlled comparison of GA, SA, and PSO. A working simulation was necessary because each optimizer could only be compared by passing its proposed service plans through the same passenger-queue model and measuring the resulting waiting time and backlog. This midterm baseline implementation therefore supported the evidence-based selection of GA; it was not intended to complete the final four-scenario experiment in advance. The complete visualization package will be produced only after the three stress scenarios are run with the same frozen model, scenario generators, and replication rule of 30 to 50 runs. This prevents the preliminary midterm evidence from being mistaken for the completed four-scenario experiment.

5. Ethical Considerations and Project Plan

5.1 Ethical and Responsible AI Considerations

The project uses aggregated public transit records and does not contain passenger names, payment details, device identifiers, or personal travel histories. Privacy risk is therefore low. The more important ethical risks are representational bias, model simplification, and overinterpretation. MTA states that APC coverage is incomplete, so recorded activity may not represent all passenger demand. A stop or period with weaker APC coverage could appear to have lower demand than it really had. The group therefore does not treat missing source rows as zero and does not describe the dataset as complete ridership.

The selected AI component is an optimizer, not an autonomous decision maker. It proposes candidate hourly service allocations inside explicit constraints, and the discrete-event simulation evaluates them. The group will label all optimized plans as decision-support results that require human review. NIST AI RMF 1.0 emphasizes managing AI risks and incorporating trustworthiness considerations throughout design, development, use, and evaluation (Tabassi, 2023). In this project, that means documenting data limits, keeping optimization constraints visible, reporting uncertainty, preserving reproducibility, and avoiding claims that the AI has discovered a real operational optimum for MTA.

Fair interpretation also requires reporting more than mean waiting time. A plan that improves the average but increases final backlog or peak queue at a particular stop may create a worse experience for some passengers. For this reason, backlog is a feasibility safeguard and P95 wait, peak queue, and occupancy are always reported as secondary metrics. The project does not use demographic attributes, so it cannot make claims about equity impacts across protected or vulnerable groups. Any future operational use would need additional equity and accessibility data.

5.2 Project Limitations

Table 18. Main project limitations

Limitation	Effect on Interpretation
2025 demand and 2026 schedule are from different periods.	The model is a controlled sensitivity analysis, not a reconstruction of a specific day.
Incomplete APC coverage and source-absent records.	Boardings and alightings are APC-observed activity and may understate total demand.
No ground-truth queue or waiting-time data.	Simulated queue and wait outputs cannot be externally validated against measured M66 queues in the current project.
Historical Boardings measure successful boardings, not all people who wanted to board.	True latent demand may be higher when passengers were passed by or chose another option.
Fixed 60-passenger capacity.	Vehicle-specific capacities are not available, so capacity-related backlog is sensitive to this assumption.
Uniform within-hour passenger arrivals.	Short within-hour bursts and schedule-aware arrival behavior are not represented.
Reference buses follow the published GTFS times exactly.	Real delay, early running, bunching, cancellation, and dwell-time variability are not represented.
Representative initial occupancy and alighting allocation.	Trip-level load and trip-level alighting patterns are approximated from aggregate stop-hour data.
Candidate schedules use even spacing and median downstream offsets.	AI improvements may reflect headway regularization as well as hourly trip reallocation.
Short-turn M660028 trips are excluded.	Service available specifically at E 68 St/Lexington Ave may be understated because the short-turn pattern begins there.
Only five eastbound stops and two weekday periods are modeled.	Findings cannot be generalized to the full M66 route, westbound service, weekends, or other routes without new analysis.
Midterm optimizer search budget is small.	GA selection is supported by controlled screening but is not proof of global optimality. Final scenario runs should be treated as near-optimal decision-support results.

5.3 Feasibility

Feasibility is supported by the completed implementation evidence summarized in Table 19. The cleaned data pipeline, verified GTFS mapping, event-driven baseline, saved passenger streams, replication checks, candidate schedule generator, common optimizer evaluator, controlled technique screening, and preliminary GA validation are complete. The remaining work applies these verified modules to the three stress scenarios, robustness checks, and the consistent final reporting package.

Table 19. Project feasibility and implementation status

Component	Midterm Status
Cleaned selected-stop ridership dataset	Complete and reproducible.
Upstream physical-stop mapping and cleaning	Complete.
Current weekday full-route GTFS reference	Complete.
Passenger stream generation and saved 30 seed sets	Complete.
Morning and Evening baseline simulation implementation	Complete and validated.
Baseline midterm validation and AI-screening outputs	Complete.
30-replication precision check	Complete; all baseline checks passed at n = 30.
GA, SA, PSO technique screening	Complete.
Final AI technique selection	GA selected.
Full baseline visualization/reporting package	Deferred to final four-scenario analysis for consistent reporting.
High-demand, reduced-service, combined scenario generators	Remaining final-phase work.
Final scenario comparison, sensitivity analysis, and report	Remaining final-phase work.

5.4 Team Responsibilities

Table 20. Team responsibilities for the Midterm Requirement

Member	Midterm Requirement Responsibilities
Elijah Crisehea M. Nollen	Experimental and Evaluation Plan, including the Python coding; Dataset Analysis; quality assurance (QA).
Ma. Sofia Anne C. Revilla	System, Simulation, and AI Design.
Karol Joy M. Ronquillo	Problem and Objectives; Ethical Considerations and Project Plan.

5.5 Timeline Until Final Submission

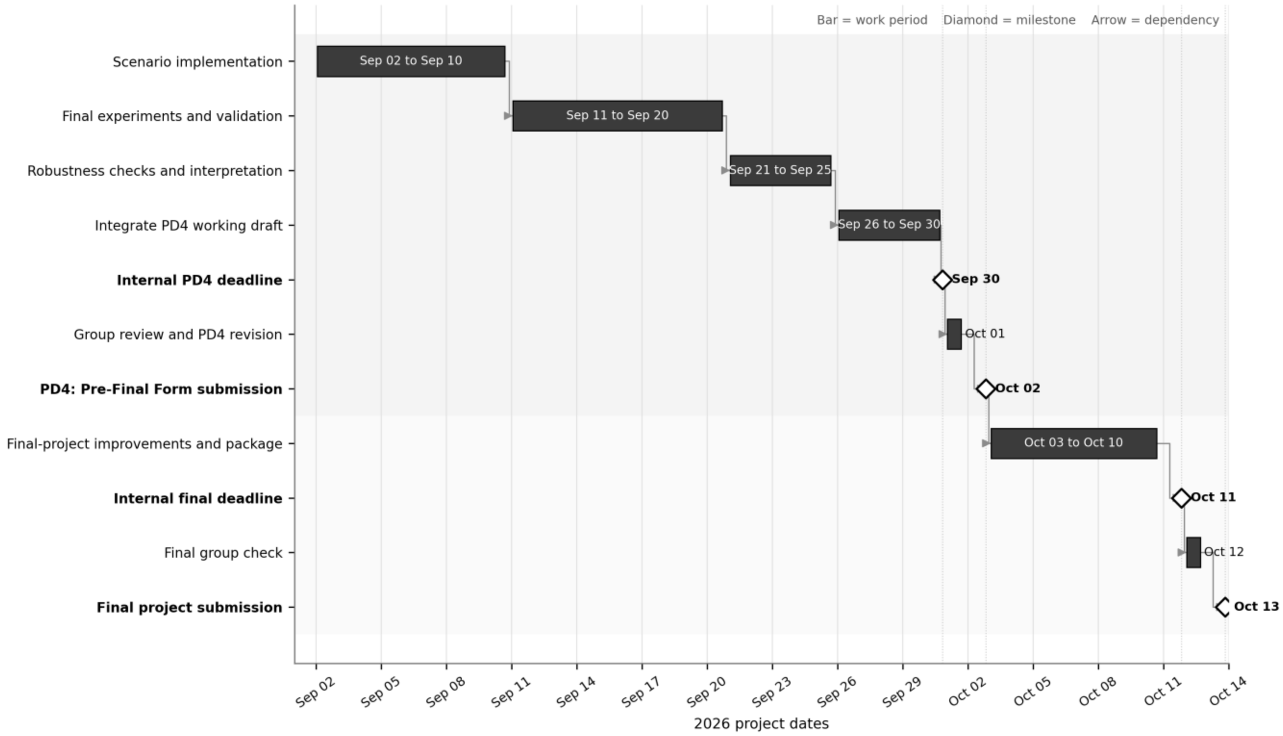

Figure 10. Gantt chart for remaining project work and submission milestones

Figure 10 shows the remaining work in dependency order, from scenario implementation through final experiments, robustness checks, PD4 integration, internal review, and the two submission milestones. Bars represent work periods, while milestone markers identify the September 30 internal PD4 deadline, October 2 Pre-Final Form submission, October 11 internal final deadline, and October 13 final submission.

6. AI Use Disclosure

6.1 Tools and Purpose of Use

The group used Google Gemini, Claude and ChatGPT during project development. AI tools were used for brainstorming, clarification of technical concepts, code-syntax guidance, grammar improvement, and drafting support. AI tools were not treated as authoritative sources. Dataset facts were checked against MTA documentation and project files, technical code behavior was checked by running the notebooks, and research claims were supported by cited publications or official documentation.

The final AI role, simulation assumptions, dataset filters, service constraints, and technique selection were based on the group's project requirements, Ms. Roselle’s feedback, cited technical references, and results from the implemented simulation and optimization experiments. The group initially considered a prediction-oriented AI role, but feedback from Ms. Roselle Gardon on the Initial Project System Representation led the members to reconsider which AI role most directly supported the simulation question. The final service-optimization design was then developed and evaluated using project-specific constraints and the same discrete-event simulation engine.

6.2 AI Prompt Log

Table 21. AI use disclosure prompt log

AI Tool	Prompt	Prompt Link	Status	Justification
Google Gemini	Convert date and hour variables python code.	Link	Modified	The AI-generated code was used only as a starting point and reference. The final code was modified to match the actual M66 project, including the variable and DataFrame names, column names, filtering conditions, file paths, stop sequences, study hours, GTFS values, and visualization structure used in the analysis.
Google Gemini	Give me Python code to convert a timestamp column into hour-of-day values.	Link	Modified	This prompt started an AI-assisted coding discussion that served as a reference for developing the baseline simulation and AI-technique experiment. The suggestions helped understand possible Python structures and methods before adapting them to the actual M66 data, simulation rules, and experimental design. The final code was modified, executed, and validated.
Claude (Anthropic)	Spot check my part of the deliverable (system simulation, AI design, and AI use disclosure) against our reference. Check whether the state variables, assumptions, and AI part line up, and flag anything missing or unclear. Do not rewrite it, only highlight what needs improvement.	Link	Modified	Claude was used to review the draft and point out gaps and mismatches in the state variables, assumptions, AI design, and AI use disclosure. The comments were used only as guidance. No generated text was copied into the paper. The group read the feedback, chose which points to apply, and made all the edits, so the review was accepted as advice while the group kept full control of the final content.

AI Tool	Prompt	Prompt Link	Status	Justification
Claude	based on this highlight the objective and problem statement that need refinement, no need to change anything just tell me where to be much more specific and align to our objective and problem statement	Link	Modified	Claude was used to analyze the draft and highlight areas in the problem statement and objective that require further specificity and alignment. The output was strictly limited to analytical feedback identifying potential gaps, without rewriting or modifying any existing text. The group evaluated the suggestions independently and manually made all refinements, retaining full authorship and control over the final content.
Claude	I'm working on a school project simulating passenger queues and wait times at 5 stops on the M66 bus route using public MTA data. Can you help me explain, in simple terms, what public datasets I'm using and why, and suggest a couple of other approaches I could have taken instead?	Link	Modified	The explanation and suggested alternatives were a good starting point, but we edited the wording and details, such as the dataset limitations and the reasoning for each alternative, to match what our group actually decided.
Gemini	I'm writing up a group project on simulating bus queues for the M66 route. Here's my draft Problem Statement and Objective, can you check it for clarity and grammar without changing what it actually says? (I have attached the draft in the prompt)	Link	Accepted	The suggested edits improved sentence clarity and fixed grammar issues while keeping the technical content accurate, so they were accepted as written.

AI Tool	Prompt	Prompt Link	Status	Justification
ChatGPT	Shortened version of the original prompt: Reorganize the Simulation Logic into 8.1 Data Preparation and Demand Modeling and 8.2 Actual Simulation Logic. Keep the existing scenarios, event rules, outputs, and replication procedures while avoiding repetition.	Link	Modified	The AI output was used as a guide, but the group revised the dataset hours, rounding rules, initial occupancy, reduced-service procedure, peak-queue update, and event-queue explanation to match the paper.

6.3 AI Responsibility Statement

The group remains responsible for selecting and correctly using the datasets, defining the project scope, writing and running the simulation code, validating the generated outputs, choosing the AI optimization technique, interpreting the results, and checking the accuracy of the final document. AI-generated text or code suggestions do not transfer responsibility for the submitted work.

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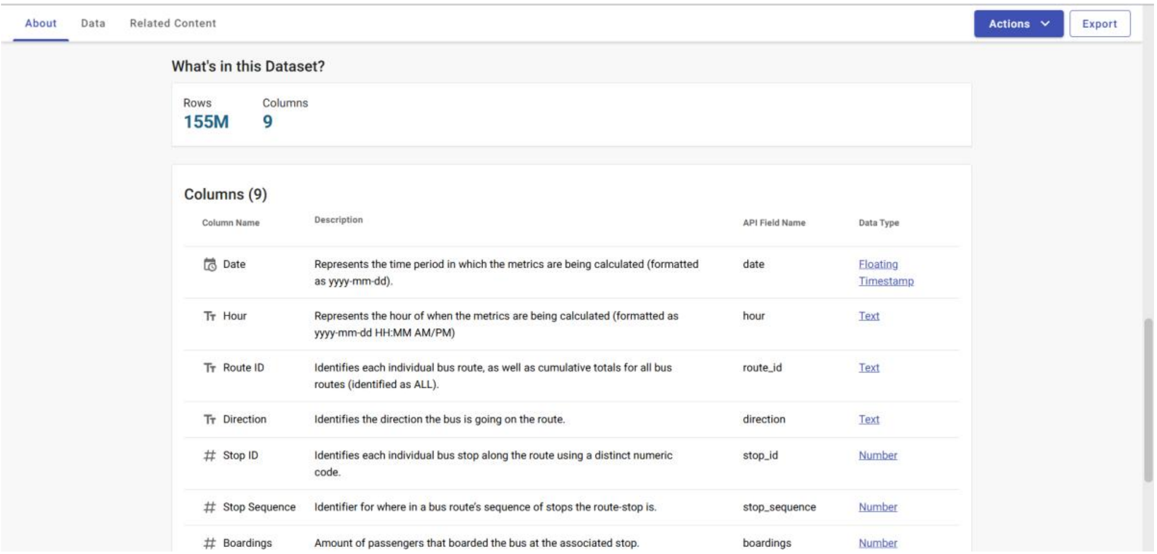
Appendix A. Dataset Acquisition and Filtering Evidence

The following screenshots were captured during the source-data acquisition process and are included as supporting reproducibility evidence. They are retained in their original captured colors so the appendix matches the source evidence used by the group.


About Data Related Content Actions Export

What's in this Dataset?

Rows 155M Columns 9

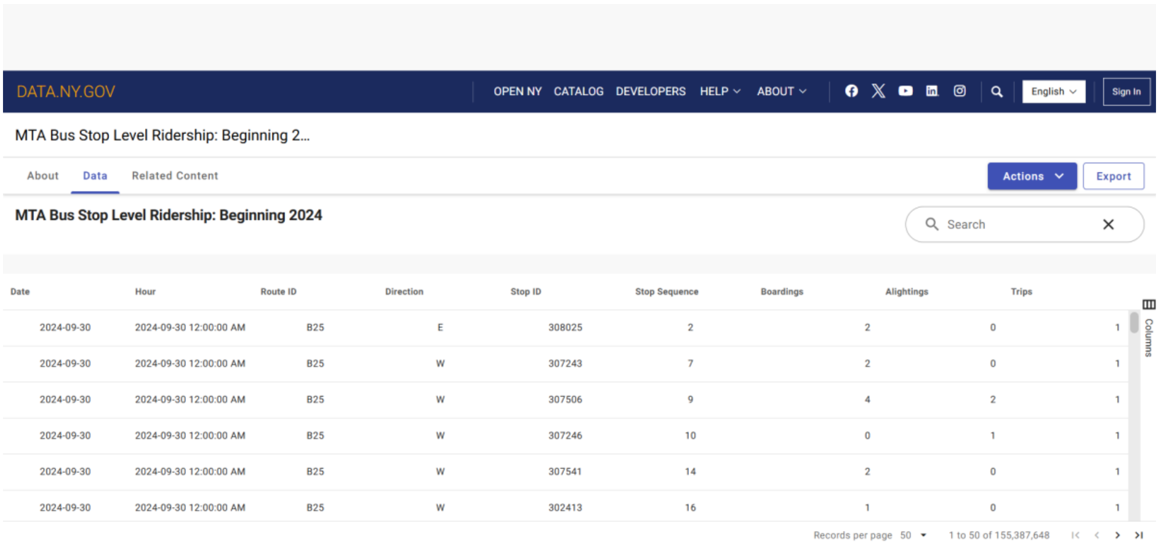
Columns (9)

Column Name	Description	API Field Name	Data Type
Date	Represents the time period in which the metrics are being calculated (formatted as yyyy-mm-dd).	date	Floating Timestamp
Hour	Represents the hour of when the metrics are being calculated (formatted as yyyy-mm-dd HH:MM AM/PM)	hour	Text
Route ID	Identifies each individual bus route, as well as cumulative totals for all bus routes (identified as ALL).	route_id	Text
Direction	Identifies the direction the bus is going on the route.	direction	Text
Stop ID	Identifies each individual bus stop along the route using a distinct numeric code.	stop_id	Number
Stop Sequence	Identifier for where in a bus route's sequence of stops the route-stop is.	stop_sequence	Number
Boardings	Amount of passengers that boarded the bus at the associated stop.	boardings	Number

Figure A1. MTA dataset overview before project-specific filtering

Note. The source overview showed approximately 155 million rows and nine variables.

This screenshot establishes the source dataset and its approximate scale before any project-specific filter was applied.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MTA Bus Stop Level Ridership: Beginning 2...

About Data Related Content Actions Export

MTA Bus Stop Level Ridership: Beginning 2024

Search

Date	Hour	Route ID	Direction	Stop ID	Stop Sequence	Boardings	Alightings	Trips
2024-09-30	2024-09-30 12:00:00 AM	B25	E	308025	2	2	0	1
2024-09-30	2024-09-30 12:00:00 AM	B25	W	307243	7	2	0	1
2024-09-30	2024-09-30 12:00:00 AM	B25	W	307506	9	4	2	1
2024-09-30	2024-09-30 12:00:00 AM	B25	W	307246	10	0	1	1
2024-09-30	2024-09-30 12:00:00 AM	B25	W	307541	14	2	0	1
2024-09-30	2024-09-30 12:00:00 AM	B25	W	302413	16	1	0	1

Columns

Records per page 50 1 to 50 of 155,387,648

Figure A2. Unfiltered MTA table with exact record count

Note. The live table showed 155,387,648 records at acquisition.

This screenshot records the exact unfiltered row count used as the starting point of the project filtering log.

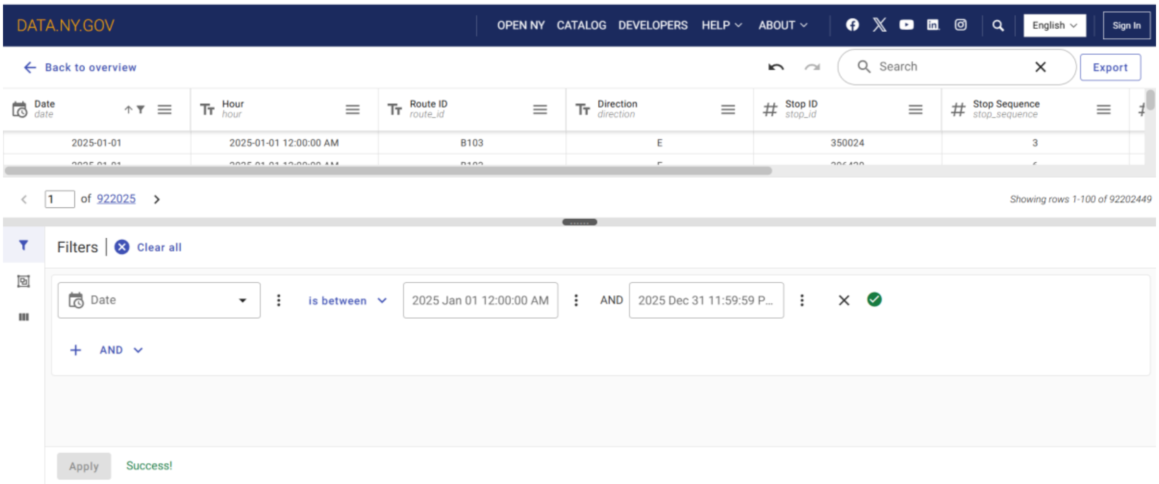

Figure A3. Calendar year 2025 filter

Note. The 2025 date filter retained 92,202,449 records.

This screenshot documents the first website-level filter, limiting the live MTA table to records from January 1 through December 31, 2025.

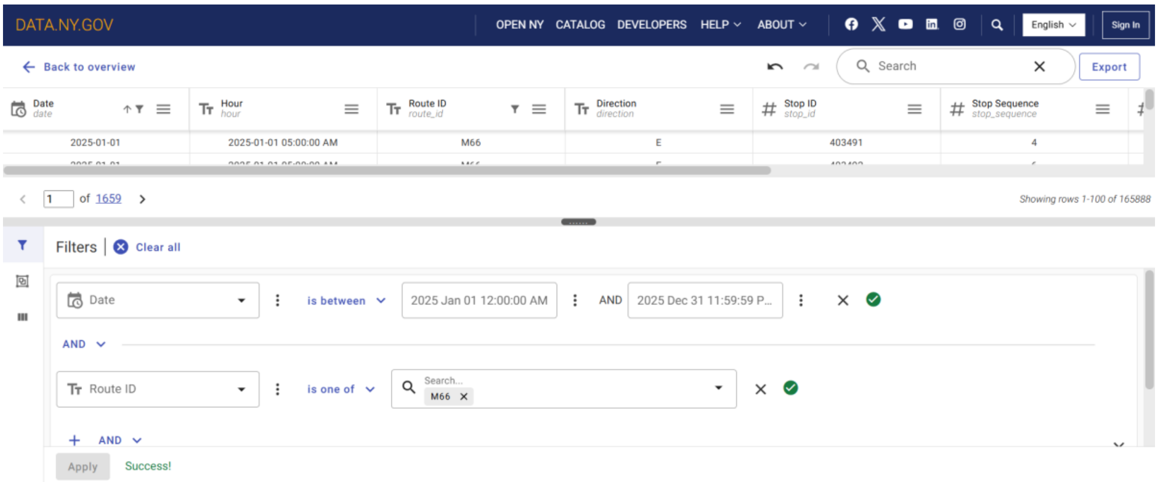

Figure A4. Calendar year 2025 and Route ID M66 filters

Note. Adding Route ID M66 retained 165,888 records.

This screenshot documents the second website-level filter, restricting the 2025 records to Route ID M66.

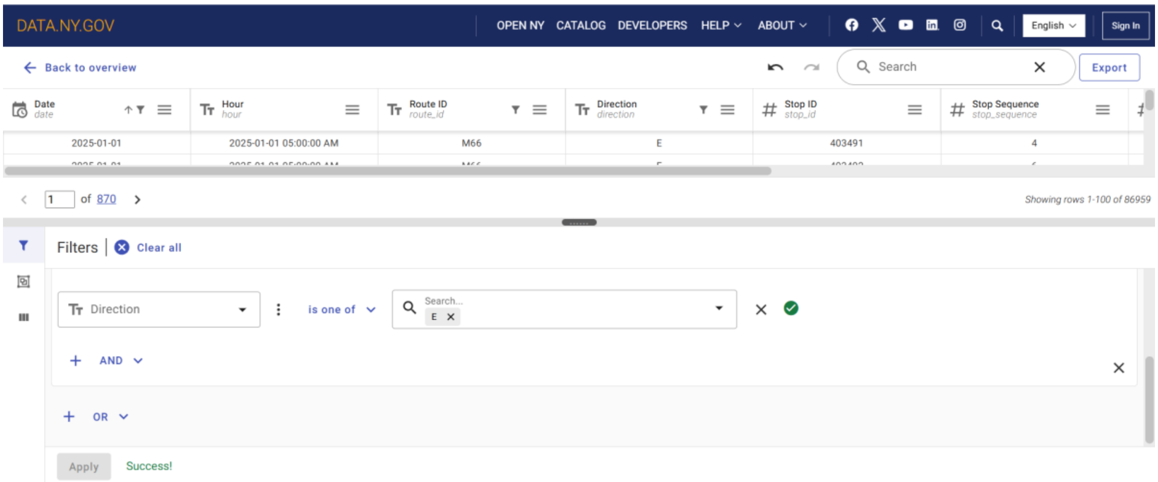

Figure A5. Final website filter for 2025 M66 eastbound records

Note. Adding Direction E retained 86,959 records, which became the raw project extract.

This screenshot documents the final website-level filter, adding eastbound Direction E. The resulting 86,959 records became the raw project extract that was later filtered and cleaned in Python.

Appendix B. Reproducibility Artifacts and Technical Documentation

The completed project package contains the notebooks, source files, processed datasets, result tables, and figures used for this paper. The most important files are listed below so that the analysis can be reproduced and checked.

GitHub repository: https://github.com/elijahnollen/m66-passenger-queue-simulation

Table B1. Main reproducibility artifacts

Artifact	Purpose
code/m66_dataset_analysis.ipynb	Reproducible dataset filtering, quality checks, statistics, mappings, and visualizations.
code/m66_baseline_simulation.ipynb	Baseline DES, replication logic, candidate schedule generator, GA-SA-PSO screening, selected AI plan evaluation, and export of results.
data/processed/MTA_M66_Selected_Stops_2025_CLEAN.csv	Final 12,814-row cleaned five-stop dataset.
data/processed/M66_Master_Passenger_Streams_30_Replications.csv	All 23,280 preserved passenger-event records for the 30 replications.
data/processed/M66_Baseline_30_Replications.csv	60 baseline period-level results: 30 Morning and 30 Evening.
data/processed/M66_Baseline_Precision_Check.csv	Baseline confidence-interval precision checks.
outputs/tables/ai_technique_run_results.csv	GA, SA, and PSO run-level screening and validation results.
outputs/tables/selected_ai_service_plans.csv	Selected GA service plans and validated metrics.

Artifact	Purpose
outputs/tables/optimized_vs_gtfs_paired_comparison.csv	Paired 30-replication comparison between the published GTFS reference and selected normal-demand GA plans.

Technical documentation used during implementation includes the official pandas documentation (https://pandas.pydata.org/docs/), NumPy documentation and SeedSequence spawning guidance (https://numpy.org/doc/), SciPy statistics documentation (https://docs.scipy.org/doc/scipy/reference/stats.html), Matplotlib documentation (https://matplotlib.org/stable/), Python heapq documentation (https://docs.python.org/3/library/heapq.html), MTA Developer Resources, and the GTFS Schedule Reference.

Appendix C. Midterm AI Technique Screening Parameters

The following settings document the controlled midterm AI comparison. They are intentionally simple because the purpose is technique selection, not exhaustive hyperparameter tuning. All techniques receive the same unique-plan evaluation budget.

Table C1. Controlled AI screening settings

Technique	Key Midterm Parameters	Common Controls
Genetic Algorithm	Population size 6; two elites; tournament selection up to 3 candidates; uniform crossover mask; 0.5 probability of neighbor mutation.	10 unique service plans per optimizer run; optimizer seeds 101, 202, 303; first 5 matched simulation replications during search; best plan validated on all 30.
Simulated Annealing	Initial temperature 0.20; cooling rate 0.93; neighbor service-plan moves; probabilistic acceptance of worse feasible plans.	Same evaluation budget, optimizer seeds, search replications, final 30-replication validation, constraints, and fitness.
Particle Swarm Optimization	Swarm size 6; inertia 0.70; cognitive coefficient 1.40; social coefficient 1.40; repaired integer service plans.	Same evaluation budget, optimizer seeds, search replications, final 30-replication validation, constraints, and fitness.

Because the search budget is small, these settings support a fair and feasible midterm comparison but do not prove that any technique reached a global optimum. The final report will describe the selected GA results as near-optimal simulation-based decision support under the tested search settings.